

CareFlow

Tessa Esman, Mara Evans,
Manuel Gardea, Zakaria Hoque,
Logan Templin

Project Description

This project focuses on developing a healthcare management system that simplifies operations across multiple medical locations.

The system brings key functions into one secure platform, including:

- Clinic and location management
- Staff and patient information
- Billing, payments, and medical records

By replacing fragmented and paper-based workflows, the application helps improve efficiency, data accuracy, and accessibility for healthcare providers.

The system supports role-based access, scalable multi-location use, and secure handling of sensitive patient data, making it suitable for small clinics, private practices, and healthcare organizations.



Target Audience

CareFlow is designed for:

- Small medical clinics
- Private medical practices
- Multi-location healthcare organizations



Primary users include:

- Administrators
- Billing staff
- Records staff
- Patient users

The system is useful for organizations looking to move away from manual or paper-based workflows.

Revenue Streams

CareFlow uses a subscription-based revenue model:

- Monthly or yearly subscription plans
- Tiered pricing based on:
  - Number of staff users
  - Number of patients supported



Optional add-ons:

- Expanded storage
- Advanced reporting
- Enhanced billing tools

This allows organizations of different sizes to choose a plan that fits their needs.

Similar Existing System

Similar Existing System: CPR

- CPR (Computerized Patient Records) is a system used to replace paper charts.
- It focuses on capturing, storing, and retrieving patient health information across care networks.
- CPR systems mainly support patient record management and basic data access.

How Our System Is Different

- Our healthcare management system is designed as a full operational platform, not just a record system.
- In addition to patient records, it supports:
- Multi-location clinic management
- Staff scheduling and assignment
- Billing and payment processing
- Secure document management

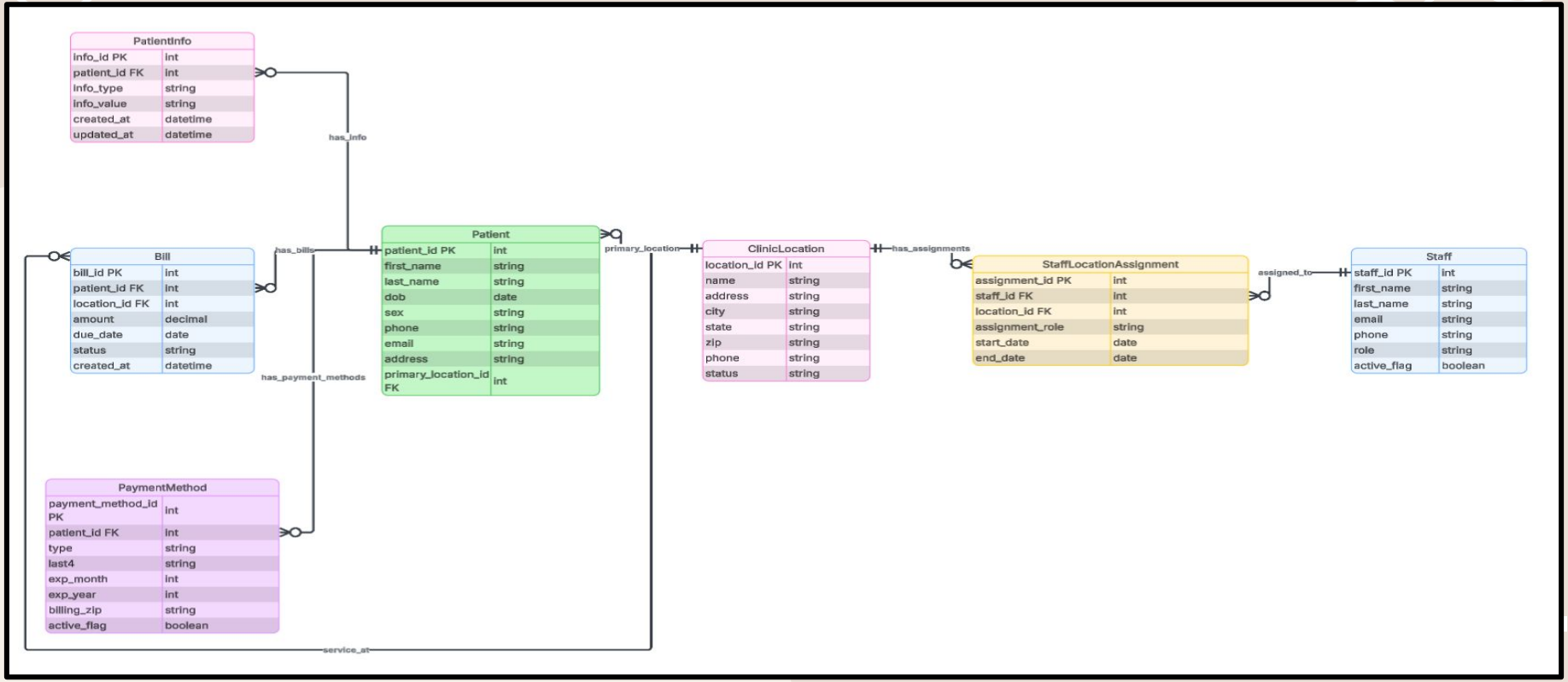
Types of Users

Administrators
- Full access to all system features
  - Managing locations
  - Managing staff
  - System configuration

Billing
- Limited to billing related functions
  - Viewing/processing payments and payment methods

Records
- Manage medical documents and patient records
  - Removing records, updating staff information

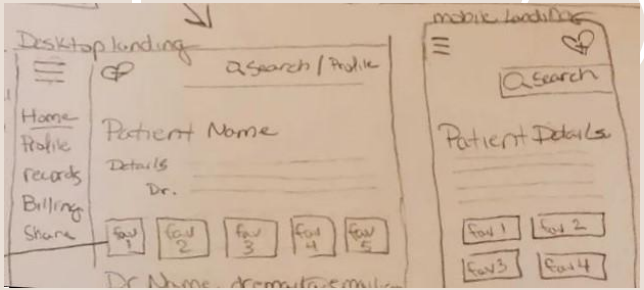
Patients
- Responsible for patient information
  - Adding, updating, maintaining this information

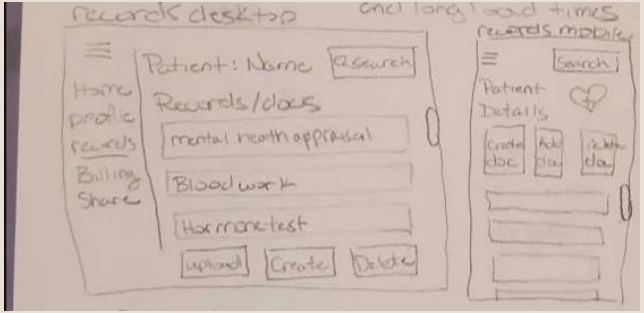

PatientInfo
info_id PK int
patient_id FK int
info_type string
info_value string
created_at datetime
updated_at datetime
has_info
Bill
bill_id PK int
patient_id FK int
location_id FK int
amount decimal
due_date date
status string
created_at datetime
has_bills
has_payment_methods
Patient
patient_id PK int
first_name string
last_name string
dob date
sex string
phone string
email string
address string
primary_location_id FK int
primary_location
ClinicLocation
location_id PK int
name string
address string
city string
state string
zip string
phone string
status string
has_assignments
StaffLocationAssignment
assignment_id PK int
staff_id FK int
location_id FK int
assignment_role string
start_date date
end_date date
assigned_to
Staff
staff_id PK int
first_name string
last_name string
email string
phone string
role string
active_flag boolean
PaymentMethod
payment_method_id PK int
patient_id FK int
type string
last4 string
exp_month int
exp_year int
billing_zip string
active_flag boolean
service_at

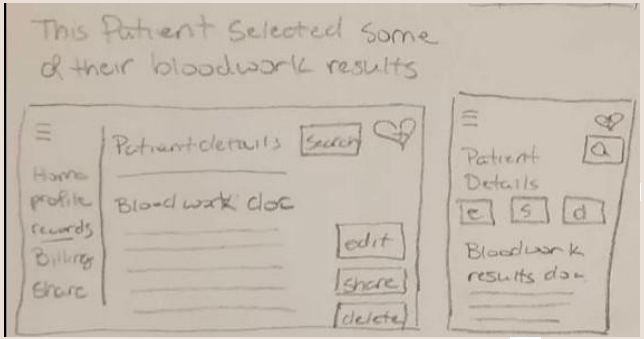
Wireframes

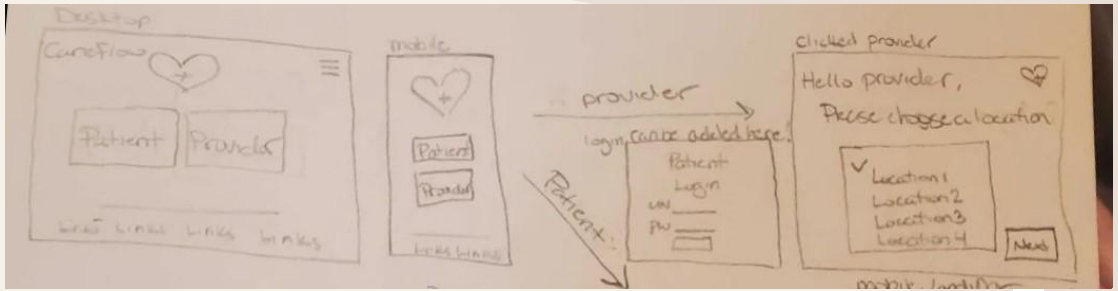
These wireframes cover the core functionality and primary user workflows for both patient and provider roles.

For the patient role, they illustrate dashboard access, profile management, viewing and updating records, and document sharing.









Wireframes

For the provider role, they show dashboard access, clinic location selection, patient management, and adding new patients.

Both desktop and mobile versions are included for each use case, ensuring that the design is responsive.

